Smoke Exhaust Scenario Lab

User Manual for Data Center Smoke Exhaust Calculations

This manual explains how to create, review, document, import, export, and report smoke exhaust calculation scenarios in the app. It is written for a first-time user and emphasizes inputs that commonly cause rejected calculations.

Important: the app is a calculation and documentation aid. It does not replace a qualified fire protection engineer, the authority having jurisdiction, insurer criteria, NFPA 92 review, or project-specific design fire approval.

Manual version	2026-07-05
App screen	Smoke Exhaust Scenario Lab
Primary output	Auditable report preview, printable report, and Excel Mode worksheet
Default reviewer mode	Predictive draft answers requiring evidence before app-ready status

1. Quick Start

Use this path when you are making your first scenario. The goal is to produce a calculation that is understandable, repeatable, and reviewable.

1. Open the app and pick the closest starting scenario in the left rail.
2. Rename the scenario so the name describes the fire case, such as MAC II packaging transient or MAC I Li-ion BBU test case.
3. Set the TSFPEWG Path inputs: Mission, MAC, Facility, Operation, Clean agent, and Direct exterior door.
4. Choose the Design Fire preset. Confirm or edit HRR, convective fraction, and ambient temperature.
5. Fill in Room and Exhaust geometry: length, width, height, smoke interface, margin, inlet count, inlet length/width, and location factor.
6. Fill in Makeup Air and Leakage: makeup ratio, free area, leakage area, flow coefficient, pressure target, and velocity limit.
7. Use Excel Mode when you want a compact spreadsheet-style view of the fire, plume, inlet geometry, plugholing, and app-added pressure checks.
8. Review the Pressure Helper. If pressure fails, use the required imbalance, max makeup ratio, and max leakage area to adjust the design.
9. Review Data Quality. Resolve Blocked issues before relying on outputs.
10. Complete Reviewer Basis. Predictive answers are drafts; replace or mark them reviewed only when you have evidence.
11. Check the Report Preview. It should show version/source control, decision path, TSFPEWG requirement trace, submittal readiness gate, formulas, variables, calculations, reviewer basis, and warnings.
12. Print the report or export scenario JSON for recordkeeping.

Successful calculation checklist: Data Quality is Ready or intentionally left Review with documented unresolved items; pressure, makeup velocity, and plugholing status are understood; reviewer basis includes sources or evidence IDs; and the Submittal Readiness Gate lists no unresolved app-blocking items.

Why the starter scenarios show Review

The starter scenarios are intentionally conservative practice cases. They usually open in Review because the app has created predictive reviewer-basis text, but a project reviewer has not yet accepted the sources, evidence IDs, pressure basis, or confirmation toggles. A label such as Rev 1 - Review 17 means revision 1 has 17 open review items. It does not mean the arithmetic failed.

What you see	What it usually means	First action
Review with many warnings	Inputs can calculate, but evidence and assumptions still need review.	Open Data Quality and resolve the listed reviewer-basis and pressure items.
Blocked	At least one required input is missing, impossible, or likely entered in the wrong units.	Fix the blocking issue before relying on any calculated output.
Ready	No active blocking errors or review warnings remain.	Still complete licensed FPE/AHJ/insurer review before issuing the calculation.

2. Screen Layout

Area	What it does	What to watch
Scenario list	Shows all scenarios and status: Ready, Review, or Blocked.	A Blocked scenario has missing or invalid information.
Scenario Data	Export, import, or load a JSON template.	Import blocks likely unit mistakes and missing critical values.
Full / Excel mode	Switches between the full submittal workflow and a compact Excel-style calculation worksheet.	Excel Mode hides reviewer-basis inputs; use Full mode before issuing a submittal.
Scenario Inputs	All editable input fields.	Numbers update calculations immediately.
Calculation View	Live visual and checks for plugholing, makeup velocity, pressure, data quality, and pressure helper.	Use this area while adjusting geometry and airflows.
Excel Mode	Simplified worksheet with input cells, Data Quality, output cells, pass/fail cells, formulas, substitutions, and app-added checks.	Use it for quick review against spreadsheet calculations, then return to Full mode before submittal.
Report Preview	Auditable output containing version/source control, TSFPEWG traceability, readiness gate, variables, formulas, trace, reviewer basis, and raw JSON.	This is what a reviewer will use to follow the calculation path.
Help manual	Opens this PDF.	Use it before importing or changing reviewer status.

3. Scenario Workflow

Use Full mode or Excel Mode

Mode	Use when	Includes
Full	Preparing a reviewer-ready package or resolving assumptions.	All scenario inputs, TSFPEWG path, reviewer basis, live visual checks, data quality, and full report preview.
Excel	Checking arithmetic quickly or comparing to the original spreadsheet calculator.	Fire/layer inputs, exhaust pickup inputs, compact Data Quality, plume and plugholing formulas, substitutions, pass/fail cells, makeup velocity limit, and app-added pressure checks.

Excel Mode is intentionally simpler, but it is not a reviewer-basis substitute. The compact Data Quality block lists blocking errors before warnings. If it shows Blocked or Review, use Full mode to resolve the source, basis, confirmation, and pressure items before issuing the calculation.

Create or duplicate scenarios

Use Duplicate when you want to test a related case. The copy keeps the original values and records the original scenario as its comparison parent. Change one major assumption at a time when possible, such as design fire, smoke interface, makeup ratio, or leakage area.

Interpret scenario status

Status	Meaning	Typical cause
Ready	No blocking errors or review warnings are active.	Inputs are valid and reviewer basis is reviewed/accepted with evidence.

Status	Meaning	Typical cause
Review	The calculation can run but contains assumptions needing review.	Predictive reviewer answers, pressure target not met, or design fire warning.
Blocked	Do not rely on outputs until fixed.	Missing HRR, bad units, smoke interface above ceiling, invalid leakage, or missing required source.

When Data Quality is Blocked, use summary values only as clues. Do not copy Required exhaust, Pressure estimate, Inlets required, or Pressure Helper values into a report until the blocking issue is fixed. For example, a smoke interface above the ceiling can create very large exhaust values that are not a valid design result.

4. Critical Input Parameters

TSFPEWG Path inputs

Input	Common options	Meaning	Impact
Mission	Mission-essential, Mission-critical	Importance of continued operation and risk tolerance.	Helps choose the chapter/path shown in Decision Path.
MAC	MAC III, MAC II, MAC I	Mission assurance category. MAC I is generally most critical.	Changes the decision path and expected reviewer rigor.
Facility	Typical, Remote, Modular	Facility configuration.	Remote or modular selections route to modified TSFPEWG paths.
Operation	Occupied, Unoccupied	Whether people are normally present.	Document occupant assumptions in reviewer basis.
Clean agent	Yes, No	Whether clean-agent protection affects exhaust operation.	If Yes, activation sequence must document manual/exhaust interlocks.
Direct exterior door	Yes, No	Whether exterior access affects smoke exhaust provisions.	For some MAC III cases, portable fan path may be considered; document exact basis.

Design Fire inputs

Input	Common value/options	Meaning	Impact
Preset	10 kW detection, 23 kW small electronic, 200 kW enclosure, 500 kW packaging, 300/600 kW electrical, Li-ion/project-specific, custom	Starting fire scenario and evidence family.	Sets default HRR and source notes; changing preset resets reviewer basis to predictive draft.
Project basis / justification	Project narrative	Why this fire applies to this room and fuel package.	Reviewer may reject calculation without this.
HRR	10, 23, 200, 300, 500, 600 kW or project-specific	Heat release rate used by plume equations.	Drives smoke mass flow and exhaust volume.
Convective fraction	Usually decimal such as 0.70	Fraction of HRR assumed convective.	The app blocks percent-style entry like 70.
Ambient temp	Often 20-25 deg C	Initial air temperature.	Affects smoke density and plugholing temperature ratio.

Do not use the 10 kW detection benchmark as an exhaust-sizing design fire unless the report clearly says it is only a detection or airflow transport sensitivity check.

Room and Exhaust inputs

Input	Common range	Meaning	Impact
Room length/width	Project dimensions in ft	Plan dimensions of the smoke zone.	Used for documentation and reviewer context.
Room height	Ceiling height in ft	Height to exhaust inlet/ceiling layer basis.	Smoke interface must be below this value.
Smoke interface	Design clear height in ft	Height from floor to smoke layer interface.	Major driver of plume mass flow and layer depth.
Exhaust margin	0% or project margin	Extra exhaust above calculated requirement.	Increases selected exhaust and can help pressure imbalance.
Selected inlets	Whole number >= 1	Number of exhaust pickup points.	Affects flow per inlet and plugholing check.
Inlet length/width	Rectangular inlet dimensions in inches	Clear exhaust inlet opening size used for equivalent diameter.	The app checks Di = 2ab/(a+b) and requires smoke depth / Di greater than 2.
Location factor	1.0 centered, 0.5 near wall where applicable	Plugholing location factor.	Lower values reduce allowed flow per inlet.

Makeup Air and Leakage inputs

Input	Common range	Meaning	Impact
Makeup ratio	Decimal such as 0.90 or 0.95	Makeup airflow divided by selected exhaust.	Higher ratio reduces pressure differential. The app blocks 90 or 95 percent-style entries.
Makeup free area	Project opening area in ft2	Net free area for incoming air.	Controls makeup velocity.
Leakage area	Project estimate/test basis in ft2	Effective leakage area through envelope/openings.	Critical pressure input. Zero or negative values are blocked.
Flow coefficient	Often around 0.60-0.70 unless justified	Discharge coefficient for leakage path.	Affects calculated pressure and required imbalance.
Pressure target	12 Pa default shown in app	Target pressure differential.	Pressure helper shows required imbalance and max makeup ratio.
Velocity limit	200 fpm default	Makeup air velocity criterion.	High velocity can disturb the smoke layer and fail the check.

5. Design Fire Presets

Preset	Default HRR	Use for	Caution
Detection benchmark	10 kW	Detector transport and airflow checks.	Not an exhaust-sizing fire.
Small electronic ignition	23 kW	Localized small electronic item sensitivity.	Not a rack or UPS fire.
Electronic enclosure/peripheral	200 kW	Moderate equipment/peripheral example.	Confirm relevance to modern listed equipment.

Preset	Default HRR	Use for	Caution
Server packaging carton	500 kW	Move-in, staging, unpacking, transient packaging.	Not normal operating server equipment.
Electrical cabinet support	300 kW	UPS/electrical support spaces.	Use only where cabinet fuel package is applicable.
Electrical cabinet upper-bound	600 kW	Severe electrical enclosure sensitivity.	Sensitivity case, not universal data-center fire.
Li-ion BBU rack	User HRR required	Battery backup rack with project data.	Requires manufacturer/UL 9540A/large-scale test basis.
Custom transient curve	User HRR required	Construction/storage/housekeeping scenarios.	Needs measured curve or documented equivalent steady HRR.

Li-ion and custom HRR workflow

Li-ion BBU rack, custom transient curve, and custom steady HRR presets do not assign a generic design fire. The app sets HRR to 0 and blocks the scenario until the user enters a project-specific equivalent steady HRR and basis.

1. Select the Li-ion or custom preset only after you have a project source such as manufacturer data, UL 9540A/large-scale test data, insurer direction, AHJ direction, or an approved engineering basis.
2. Enter the equivalent steady HRR in kW. Do not leave HRR at 0.
3. Update Project basis / justification with the test, source, fuel package, rack/cabinet configuration, and why an equivalent steady HRR is acceptable.
4. Complete Reviewer Basis fields for fuel package, design fire source/page, HRR curve basis, approval basis, equation applicability, and plume limitations.
5. Keep Data Quality in Review until the project evidence is entered and the responsible reviewer has accepted the assumptions.

6. Reviewer Basis and Predictive Answers

The app fills reviewer-basis answers as predictive drafts so first-time users are not starting from a blank page. Predictive answers are not final evidence. They explain what the app thinks the answer should cover and must be replaced, edited, or marked reviewed with sources before submittal.

Audit field	Purpose	How to use
Status	Shows whether the answer is predictive, engineer reviewed, or AHJ/insurer accepted.	Leave Predictive while drafting. Move to Engineer reviewed only when checked. Use AHJ/insurer accepted only with documented acceptance.
Source type	Classifies the evidence.	Choose hand calculation, published test, manufacturer data, AHJ direction, insurer direction, or project document when reviewed.
Source/page/test ID	Makes the answer traceable.	Enter document name, page, table, test number, calculation ID, or meeting record.
Evidence ID/link	Points to the supporting file or record.	Use a calc number, submittal ID, document control ID, or URL/path.
Confidence	Indicates strength of the basis.	Use Low for predictive assumptions, Medium for checked project basis, High for direct evidence or accepted criteria.

A scenario can be Ready only when reviewer-basis entries are reviewed or accepted and have traceable source/evidence metadata. If a reviewed item still says App prediction or PREDICTIVE-DRAFT, the app warns you.

Ready checklist for reviewer basis

Check	Required before relying on Ready status
Status	Every reviewer-basis item is Engineer reviewed or AHJ/insurer accepted.
Source type	No reviewed item still uses App prediction.
Source/page/test ID	Each reviewed item names the document, page, table, test ID, calc ID, or meeting record.
Evidence ID/link	No reviewed item still uses PREDICTIVE-DRAFT; each item points to a file, record, URL, submittal, or calculation ID.
Confirmation toggles	Axisymmetric plume, steady-equivalent basis, no wall/rack plume, and suppression interaction are addressed or intentionally left Review with explanation.

7. Running and Interpreting the Calculation

Summary strip

The top strip shows Data quality, Required exhaust, Selected exhaust, Makeup velocity, Pressure estimate, Needed imbalance, and Inlets required. Use this as a quick health check, then review the detailed sections.

Pressure Helper

Output	Meaning	How to act
Current imbalance	Selected exhaust minus makeup airflow.	If small, pressure will usually be low.
Needed imbalance	Imbalance required to hit target pressure with current leakage.	Compare against current imbalance.
Max makeup ratio	Highest makeup ratio that can meet target pressure.	If Not possible, selected exhaust/leakage/target must change.
Max leakage area	Largest leakage area that can meet target with current imbalance.	Use to test envelope/damper assumptions.

Default pressure example

In the starter Packaging transient scenario, the app may show makeup ratio 0.95, pressure near 0.1 Pa, needed imbalance about 5,150 cfm, and max makeup ratio about 0.28. This tells the user that 95 percent makeup leaves too little exhaust imbalance to meet the 12 Pa pressure target with the entered leakage area.

If the helper says	Interpretation	Typical next step
Max makeup ratio = 0.28	Makeup must be 28 percent of selected exhaust or lower for the current leakage and pressure target.	Reduce makeup, increase selected exhaust, reduce leakage, or revisit the pressure target with the reviewer.
Max makeup ratio = Not possible	Even zero makeup is not enough with current selected exhaust/leakage/target.	Increase selected exhaust, lower leakage, lower target if allowed, or change the smoke-control strategy.
Max leakage area is very small	The pressure target depends on a tight envelope or closed dampers/doors.	Tie leakage area to drawings, damper schedule, or commissioning test evidence.

Plugholing and makeup velocity

- Plugholing Fail means either the smoke depth to equivalent inlet diameter ratio is not greater than 2, or flow per inlet exceeds the prototype plugholing limit. Increase smoke layer depth, reduce inlet equivalent diameter, increase inlet count, or revisit exhaust arrangement.
- For rectangular exhaust inlets, the app calculates Di = 2ab/(a+b) from inlet length and width in inches, then checks d/Di before relying on the plugholing flow equation.
- Makeup velocity Fail means incoming air exceeds the selected velocity criterion. Increase free area or adjust makeup flow.
- Pressure Fail means the selected makeup/leakage/exhaust imbalance does not meet the pressure target. Use the Pressure Helper before changing arbitrary values.

8. Import, Export, and Templates

Use Export to copy the current scenario set as JSON. Use Template to load a sample import payload. Use Import to replace the scenario set with the pasted JSON.

Caution: Import replaces the current scenario set when the import succeeds. Export the current scenarios before importing if you may need to return to them. If an import is blocked by errors, the app keeps the last valid scenario set active and shows the first blocking issue in Scenario Data. Imports are limited to the first 25 scenarios and show a warning if additional pasted scenarios were not imported.

Import check	Why it matters
Known fields only	Unknown fields are ignored with a warning so accidental spreadsheet columns do not affect the calculation.
Expected units block	The app expects ft, ft2, in, kW, deg C, Pa, fpm, count, percent, and decimal ratio/factor values.
Decimal ratios	Makeup ratio and convective fraction must be 0.95 and 0.70 style decimals, not 95 or 70. Flow coefficient and exhaust location factor are also decimal factors.
Count and percent fields	Selected inlet count must be a whole count. Exhaust margin is a percent value such as 10 for 10%, not 0.10.
Project-specific HRR	Li-ion BBU and custom fires must include a nonzero HRR or equivalent steady basis.
Reviewer audit metadata	Import/export preserves predictive/reviewed status, source type, source reference, evidence ID, and confidence.

Import result messages

Message	Meaning	User action
Imported with warnings	The JSON was accepted, but Data Quality still has review items.	Open Data Quality and resolve warnings before relying on the scenario.
Import blocked	The JSON was not accepted because at least one error was found.	Fix the listed unit, enum, number, boolean, or missing-HRR issue and import again.
Ignored unknown field	A pasted field name is not part of the app schema.	Check spreadsheet headers or exported JSON field names.
First 25 imported	The pasted JSON had more than 25 scenarios. The app imported only the first 25.	Split the import or confirm the intended scenarios are in the first 25 records.

9. Report Output

The Report Preview is the primary audit trail. It includes Version and Source Control, Decision Path, TSFPEWG Requirement Trace, Submittal Readiness Gate, Design Fire Basis, Pressure Helper, Plugholing and Inlet Geometry, Reviewer Basis, Data Quality, Variable Register, Formula Register, Calculation Trace, Compliance Checklist, and Raw JSON.

- Decision Path explains the selected TSFPEWG route and major triggers.
- TSFPEWG Requirement Trace maps each applicable smoke exhaust requirement to the scenario input, app response, and remaining evidence need.
- Submittal Readiness Gate separates app-blocking issues from external licensed FPE/AHJ/insurer approval.
- Plugholing and Inlet Geometry shows rectangular inlet dimensions, equivalent diameter, d/Di ratio, flow per inlet, required inlet count, and separation distance.
- Variable Register lists each input or calculated variable, value, unit, source, and note.
- Formula Register lists formulas used by the app.
- Calculation Trace shows substitutions and results step by step.
- Reviewer Basis shows every predictive or reviewed answer with source and evidence metadata.
- Raw JSON preserves the exact scenario state for reproducibility.

10. Common Rejection Causes and Prevention

Rejection cause	Prevention
Design fire not justified	Complete fuel package, source/page, HRR curve basis, approval basis, and evidence ID.
Equation path not valid	Document plume applicability and limitations; confirm this is not a wall, rack, shielded, confined, or multi-source case.
TSFPEWG path too vague	Enter exact sections, edition assumptions, exceptions, and AHJ interpretations.
Leakage basis unsupported	Tie leakage area and coefficient to drawings, door/damper schedule, commissioning test, or other project evidence.
No validation	Add independent hand calculation, handbook example, benchmark case, or peer-check ID.
Unit mistake	Use app units, whole inlet counts, percent margin values, and decimal ratio/factor fields; rely on import validation before trusting imported scenarios.
Pressure target misunderstood	Use the Pressure Helper to show whether makeup reduction alone can satisfy target pressure.
Traceability missing	Use the TSFPEWG Requirement Trace to show each triggered path, calculation, control, and evidence item.

11. Validation Package

The project folder includes a validation package for product-level review. These files do not replace professional review, but they make the app easier to test and defend.

File	Purpose
validation/requirements-traceability.json	Machine-readable TSFPEWG requirement matrix used by the app report.

File	Purpose
validation/scenario-schema.json	Formal import/export field, unit, enum, and schema summary.
validation/golden-cases.json	Locked regression cases across MAC III, MAC II, MAC I, remote, modular, Li-ion, and failure paths.
validation/run-regression-tests.mjs	Automated check that golden cases, schema, and traceability still match the app engine.
validation/reviewer-package-checklist.md	Independent reviewer checklist and signoff fields.
validation/acceptance-criteria.md	Software, calculation, and external acceptance rules.
validation/excel-comparison.md	Comparison of Excel calculator scope versus app Excel Mode and app-added checks.

Run the regression check after formula, schema, report, or import changes: node validation/run-regression-tests.mjs. The golden cases are software baselines and still require independent hand-calculation/FPE signoff before being treated as certified design examples.

12. First Calculation Example

1. Select Packaging transient in the scenario list.
2. Confirm MAC II, Mission-essential, Typical facility, Occupied, no clean agent, no direct exterior door.
3. Confirm design fire preset Server packaging carton - 500 kW.
4. Use room dimensions 60 ft by 40 ft by 18 ft, smoke interface 8 ft, and exhaust inlet dimensions 24 in by 24 in.
5. Use makeup ratio 0.95, free area 260 ft2, leakage area 9 ft2, flow coefficient 0.65, and pressure target 12 Pa.
6. Read Required exhaust and Selected exhaust in the summary strip.
7. Read Pressure Helper. The default case will likely show pressure review because the makeup ratio is high relative to leakage.
8. Edit reviewer basis. Keep predictive status while drafting. Move items to Engineer reviewed only after entering source/page/test and evidence ID.
9. Print the report when the calculation and reviewer basis are ready for review.

13. Limitations

- The app implements a simplified steady axisymmetric plume calculation path and does not replace NFPA 92, TSFPEWG, AHJ, insurer, or qualified FPE review.
- Predictive reviewer-basis answers are draft prompts, not accepted engineering evidence.
- The app can report app-ready status, but project use still requires external licensed FPE, AHJ, insurer, or owner acceptance as applicable.
- Li-ion BBU, transient combustible, rack, cabinet, shielded, suppression-controlled, or multi-source cases may require project-specific testing or different modeling.
- Pressure calculations depend strongly on leakage area, coefficient, HVAC fire mode, fan sequence, wind, stack effect, damper state, and commissioning assumptions.